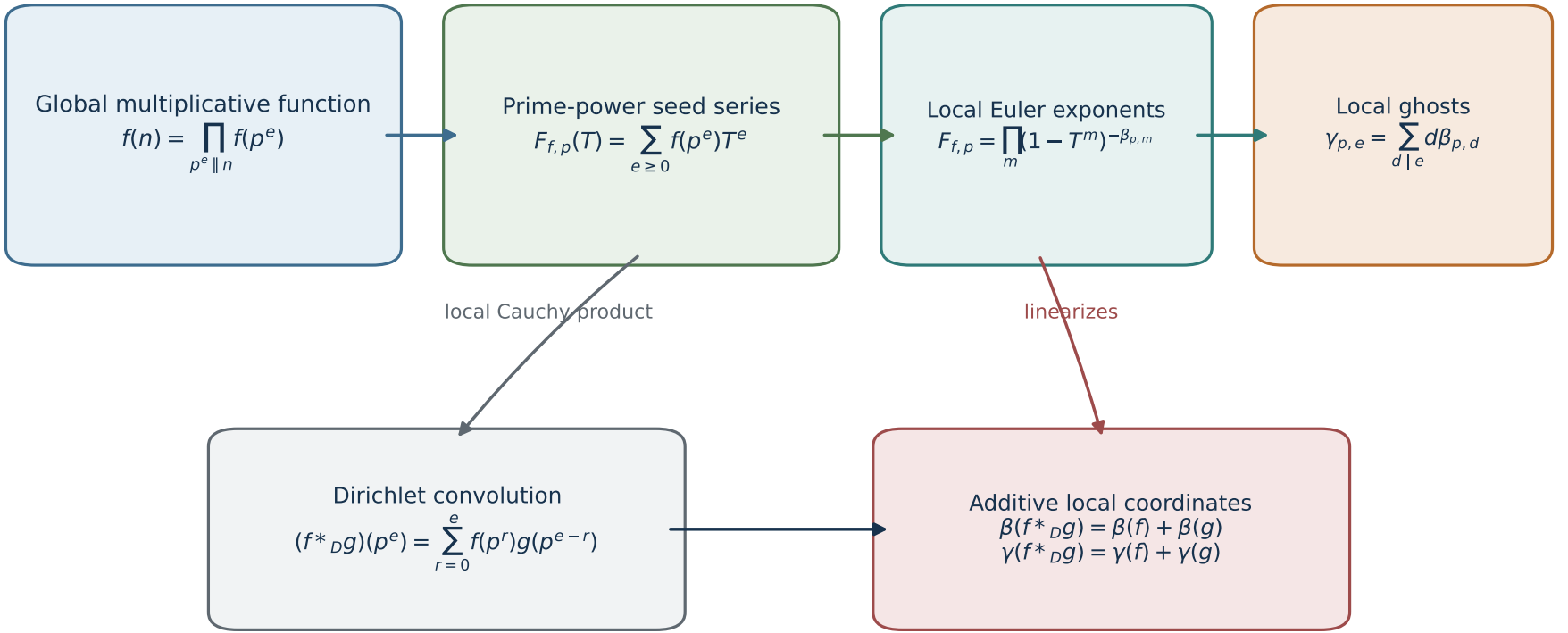
Multiplicative functions as descent from prime-power local series



The local charts glue effectively because unique factorization makes every global coefficient depend on finitely many primes.